Lesson 1: Factoring Review: Common Factors and $x^2 + bx + c$

Assignment

1. Factor by taking out the GCF.

(a) $14a^3 - 21a^2$

(b) $9x^2y - 6xy^2 + 3xy$

(c) $-8p^4 + 20p^2$ (take out a negative factor)

(d) $24m^5n^2 - 16m^3n^3 + 40m^2n^4$

2. Factor.

(a) $5x(2x + 1) - 3(2x + 1)$

(b) $y(y - 4) + (y - 4)$

(c) $7(a - 3) - a(3 - a)$

(d) $2x(x + 5)^2 - 6(x + 5)$

3. Factor by grouping.

(a) $x^3 + 2x^2 + 5x + 10$

(b) $4ab + 12a - b - 3$

(c) $2x^2 - 6x + 5x - 15$

(d) $mn - 8 + 2n - 4m$

4. Factor where possible. If a trinomial does not factor over the integers, say so.

(a) $x^2 + 8x + 15$

(b) $x^2 - 9x + 14$

(c) $a^2 + 4a - 32$

(d) $y^2 - y - 42$

(e) $n^2 - 16n + 63$

(f) $x^2 + 19x + 48$

(g) $t^2 + 2t - 63$

(h) $k^2 - 20k + 36$

(i) $x^2 + x + 3$

5. Factor.
(a) $x^2 + 9xy + 20y^2$ (b) $a^2 - 2ab - 15b^2$ (c) $m^2 - 13mn + 40n^2$ (d) $p^2 + 3pq - 28q^2$

6. Factor fully.
(a) $5x^2 + 10x - 40$ (b) $2y^3 - 22y^2 + 48y$ (c) $-x^2 - 3x + 54$

(d) $40 + 3x - x^2$ (e) $4a^2b - 4ab - 80b$ (f) $3x^2y + 3xy^2 - 18y^3$

7. Two students factored $2x^2 - 10x - 28$. Ali wrote $(2x + 4)(x - 7)$. Bree wrote $2(x - 7)(x + 2)$.
(a) Expand Ali's answer. Is it equal to $2x^2 - 10x - 28$?

(b) Whose answer is factored fully? Explain.

8. Which trinomials can be factored over the integers? Factor those that can; for the others, show why not.
(a) $x^2 + 7x + 12$ (b) $x^2 + 4x + 6$ (c) $x^2 - x - 20$ (d) $x^2 - 5x - 7$

9. (a) Find all integers b so that $x^2 + bx + 15$ can be factored.

(b) Find all positive integers k so that $x^2 + 9x + k$ can be factored.

10. Decide whether the binomial is a factor of the polynomial. Show your factoring.
(a) Is $x - 5$ a factor of $x^2 - 2x - 15$? (b) Is $x + 4$ a factor of $3x^2 + 6x - 24$?

(c) Is $x + 2$ a factor of $x^3 + 2x^2 + 3x + 6$? (d) Is $x - 3$ a factor of $x^2 + x - 6$?

11. A rectangular garden plot has an area of $(x^2 + 14x + 45)$ m^2.
(a) Factor to find expressions for the length and width.

(b) How much longer is the length than the width?

(c) Write and simplify an expression for the perimeter, then evaluate it when $x = 11$.

12. A box has a volume of $(2x^3 + 10x^2 + 12x)$ cm^3. Its height is $2x$ cm. Find expressions for the length and width of its base, and the box's dimensions when $x = 4$.

13. Which binomial is a factor of both $2x^2 - 8x - 42$ and $x^2 + x - 6$?

A. $x - 7$ **B.** $x + 3$ **C.** $x - 3$ **D.** $x + 2$

14. The trinomial $x^2 + kx - 12$ can be factored over the integers for each of these values of k **except**

A. 1 **B.** -4 **C.** 11 **D.** 6

15. One factor of $12 + 4x - x^2$ is

A. $x - 2$ **B.** $x + 6$ **C.** $x - 6$ **D.** $x - 12$

16. Numerical Response The trinomial $x^2 + bx + 12$, where b is a positive integer, can be factored over the integers for several values of b. The sum of all these values of b is _____.

Lesson 2: Factoring $ax^2 + bx + c$

Assignment

1. Factor by decomposition, or show that the trinomial does not factor over the integers.
(a) $2x^2 + 9x + 4$ (b) $3x^2 + 16x + 5$ (c) $4x^2 + 12x + 5$

(d) $6x^2 + 23x + 20$ (e) $3x^2 + 5x + 4$

2. Factor.
(a) $5x^2 - 17x + 6$ (b) $2a^2 + a - 15$ (c) $6y^2 - y - 12$ (d) $8x^2 - 14x + 3$

(e) $10m^2 + 11m - 6$ (f) $9t^2 - 9t - 4$ (g) $12x^2 - 25x + 12$ (h) $7n^2 + 2n - 9$

3. Complete the steps to factor $6x^2 - 7x - 20$ by decomposition.
$ac =$ _____; the two numbers are _____ and _____; $6x^2 - 7x - 20 = 6x^2$ _____ _____ $- 20$

4. Factor $2x^2 + 5x - 12$ by guess and check. Show at least one try that does not work.

5. A student factored $6x^2 - x - 2$ as $(3x - 2)(2x - 1)$. Check the answer by expanding. If it is wrong, find the correct factoring.

6. Factor fully.

(a) $6x^2 + 21x + 9$

(b) $20a^2 - 10a - 30$

(c) $12x^3 + 14x^2 - 6x$

(d) $8y^4 - 26y^3 + 15y^2$

7. Factor fully.

(a) $-3x^2 + 10x + 8$

(b) $6 - x - 12x^2$

(c) $-10x^2 - 13x + 3$

(d) $-4x^3 + 2x^2 + 30x$

8. Factor.

(a) $2x^2 + 7xy + 3y^2$ (b) $6a^2 - 5ab - 4b^2$ (c) $4m^2 - 12mn + 5n^2$ (d) $15x^2 + 2xy - 8y^2$

9. Factor fully by first taking out a fractional common factor.

(a) $\frac{1}{2}x^2 - 3x + 4$

(b) $\frac{1}{5}x^2 + x - 10$

(c) $\frac{2}{3}x^2 + 2x - 12$

(d) $\frac{1}{2}x^2 + \frac{7}{6}x + \frac{1}{3}$

10. Which are perfect-square trinomials? Factor those that are.

(a) $x^2 + 14x + 49$ (b) $4y^2 - 20y + 25$ (c) $9x^2 + 15x + 25$

(d) $x^2 - 12x - 36$ (e) $49a^2 - 56ab + 16b^2$ (f) $1 + 6t + 9t^2$

11. Fill in the blank so that each trinomial is a perfect square. Use a positive middle term where there is a choice.

(a) $x^2 + _____ + 36$ (b) $64x^2 - _____ + 9$ (c) $25y^2 + 70y + _____$

(d) $_____ - 36xy + 4y^2$ (e) $\frac{1}{4}x^2 + _____ + 9$

12. Factor fully.
(a) $81x^2 - 18x + 1$ (b) $4 + 28t + 49t^2$ (c) $3x^2 - 30x + 75$ (d) $16a^2 + 24ab + 9b^2$

13. A rectangular poster has an area of $(6x^2 + x - 15)$ cm^2. It is mounted in a frame that is 2 cm wide on every side.
(a) Factor to find the poster's dimensions.

(b) Write expressions for the outer length and width of the frame.

(c) Find and simplify an expression for the area of the frame alone.

(d) Find the area of the frame when $x = 10$.

14. One factor of $18x^2 - 3x - 6$ is
A. $3x + 2$ **B.** $2x - 1$ **C.** $2x + 1$ **D.** $x - 2$

15. Which trinomial is **not** a perfect square?
A. $9a^2 - 30ab + 25b^2$ **B.** $4x^2 + 12x + 9$ **C.** $x^2 + 8x + 64$ **D.** $1 - 2y + y^2$

16. If $2x^2 + kx - 15 = (2x - 3)(x + 5)$, then the value of k is
A. 7 **B.** -7 **C.** 13 **D.** 2

17. Numerical Response The trinomial $15x^2 + 14x - 8$ can be written in the form $(ax - b)(cx + d)$, where a, b, c and d are positive integers. Record a in the first box, b in the second, c in the third and d in the fourth.

18. Numerical Response The trinomial $4x^2 + bx + 25$ is a perfect square and $b > 0$. The value of b is _____.

Lesson 3: Factoring by Substitution: $a(f(x))^2 + b(f(x)) + c$

Assignment

1. Factor.

(a) $x^4 + 8x^2 + 15$

(b) $a^4 - 2a^2 - 35$

(c) $y^6 + 2y^3 - 15$

(d) $x^2y^2 + 3xy - 18$

2. Factor fully.

(a) $x^4 - 10x^2 + 9$

(b) $n^4 - 21n^2 - 100$

(c) $x^4 + 3x^2 - 4$

(d) $5x^4 - 25x^2 + 20$

(e) $x^5 - 8x^3 + 16x$

(f) $x^4 - 13x^2y^2 + 36y^4$

3. Factor fully.

(a) $2x^4 + 7x^2 + 3$

(b) $3a^4 - 7a^2 + 2$

(c) $4x^4 - 17x^2 + 4$

(d) $5m^6 + 13m^3 - 6$

(e) $6x^2y^2 - 7xy - 20$

(f) $12x^5 - 15x^3 - 27x$

4. Factor fully: $9 - 8x^2 - x^4$. (Hint: take out -1 first.)

5. Factor each trigonometric expression.

(a) $2\cos^2 x - 3\cos x + 1$

(b) $3\sin^2 x + 5\sin x - 2$

6. Factor fully. Simplify each factor.

(a) $(x+3)^2 + 5(x+3) + 6$

(b) $(a-2)^2 - 4(a-2) - 21$

(c) $3(x+1)^2 - 7(x+1) + 2$

(d) $2(2x-3)^2 + (2x-3) - 6$

(e) $4(y+1)^2 - 12(y+1) + 9$

(f) $6(1-x)^2 - 5(1-x) - 4$

(g) $(x+y)^2 - 2(x+y) - 15$

(h) $2(x+2)^2 + 2(x+2) - 12$

7. Factor $2(x-3)^2 - 3(x-3) - 2$ two ways: by substitution and by expanding first. Which way was faster?

8. Factor fully: $(x+4)^2 - 5(x+4)(x-1) + 6(x-1)^2$.

9. The area of a rectangle is $(x+5)^2 + (x+5) - 12$ square units. Its length and width are binomials in x.

(a) Factor to find expressions for the length and width.

(b) The perimeter is 42 units. Find x and the area.

10. Find all integers k for which $x^4 + kx^2 - 12$ can be factored as $(x^2+p)(x^2+q)$, where p and q are integers.

11. Which expression is a perfect square trinomial?

A. $x^4 - 6x^2 + 9$ **B.** $x^4 + 6x^2 - 9$ **C.** $4x^4 + 6x^2 + 9$ **D.** $x^4 - 9x^2 + 9$

12. When fully factored, $2(x+1)^2 - 7(x+1) + 3$ is equal to

A. $(2x-1)(x-3)$ **B.** $(2x+1)(x-2)$ **C.** $2x(x-2)$ **D.** $(2x+3)(x-2)$

13. Which expression is **not** a factor of $3x^4 + 3x^2 - 36$?

A. 3 **B.** $x^2 + 4$ **C.** $x^2 - 3$ **D.** $x - 2$

14. Numerical Response The expression $3(2x-1)^2 - 4(2x-1) - 4$ can be factored as $(ax+b)(cx+d)$, where a, b, c and d are integers and $a > c > 0$. The value of $a+b+c+d$ is _____.

15. Numerical Response When $x^4 - 29x^2 + 100$ is factored fully, it is $(x-a)(x+a)(x-b)(x+b)$, where $0 < a < b$. The value of ab is _____.

Lesson 4: Differences of Squares, Including $a^2(f(x))^2 - b^2(g(y))^2$

Assignment

1. Factor, if possible.

(a) $36x^2 - 49$ (b) $64a^2 - 25b^2$ (c) $81p^2q^2 - 4r^2$

(d) $121 - x^2y^2$ (e) $9x^2 + 16$ (f) $-25 + 4m^2$

2. Factor fully.

(a) $18x^2 - 50y^2$ (b) $7a^3 - 28a$ (c) $48m^2n - 3n^3$ (d) $20x^2 + 45$

(e) $72 - 2y^2$ (f) $-27x^2 + 12y^2$ (g) $x^3y - xy^3$ (h) $50a^4b^2 - 2a^2b^4$

3. Factor. (a) $1 - \cos^2 x$ (b) $9\sin^2 x - 4$

4. Factor fully.

(a) $\frac{1}{9}x^2 - 4$ (b) $\frac{x^2}{25} - \frac{y^2}{36}$ (c) $0.16a^2 - 0.01b^2$

(d) $\frac{4}{49}m^2 - 1$ (e) $\frac{1}{2}x^2 - 8$

5. A rectangular rug has an area of $64x^2 - 9y^2$ square metres. Its length and width are both of the form $ax + by$, where a and b are integers.

(a) Write expressions for the length and width.

(b) The perimeter is 80 m. Determine x.

(c) The length is also 6 m more than the width. Determine y, the dimensions and the area.

6. Factor fully.

(a) $a^4 - 1$

(b) $81x^4 - 16y^4$

(c) $2m^4 - 162$

(d) $x^8 - 256$

(e) $16 - 81y^4$

(f) $x^6y^2 - 4$

(g) $\frac{1}{16}x^4 - 1$

7. Factor fully. Simplify each factor.

(a) $(x-4)^2-9$

(b) $36-(a+2)^2$

(c) $(2x+1)^2-(x+3)^2$

(d) $9(a+1)^2-4a^2$

(e) $x^2-(y-z)^2$

(f) $49(x-1)^2-16(y+3)^2$

(g) $(x+y)^2-(x-y)^2$

(h) $4(3a-2)^2-25(a-1)^2$

8. A metal washer is a disc of radius $(r+4)$ mm with a hole of radius $(r-2)$ mm cut from its centre.

(a) Write, without simplifying, an expression for the area of the washer.

(b) Factor the expression fully.

(c) Use the factored form to find the exact area when $r=3$.

9. Factor fully. Choose the method using the checklist.

(a) $5x^2-20x-60$

(b) $2x^3-3x^2-8x+12$

(c) $6y^4-5y^2-6$

(d) $9x^2-12x+4$

(e) x^4-8x^2-9

(f) $a^2+10a+25-4b^2$

10. Determine whether each binomial is a factor of $2x^4 - 32$. Explain. (a) $x + 2$ (b) $x - 4$ (c) $x^2 + 4$

11. Raj factored $4x^2 - 36$ as $(2x - 6)(2x + 6)$ and said it was fully factored. Explain his error and factor fully.

12. One factor of $16x^4 - 1$ is

A. $4x - 1$ **B.** $2x^2 + 1$ **C.** $2x + 1$ **D.** $4x + 1$

13. The polynomials $6x^2 + xy - 2y^2$ and $9x^2 - 4y^2$ have a common factor of

A. $3x - 2y$ **B.** $3x + 2y$ **C.** $2x - y$ **D.** $2x + y$

14. Which binomial **cannot** be factored over the real numbers?

A. $4x^2 - 1$ **B.** $x^2 + 25$ **C.** $25 - x^2$ **D.** $x^4 - 25$

15. Numerical Response The expression $9(x + 2)^2 - 16(x - 1)^2$ can be factored as $-(x - a)(bx + c)$, where a, b and c are natural numbers. The value of $a + b + c$ is _____.

16. Numerical Response When $x^4 - 625$ is factored fully, it is $(x - a)(x + a)(x^2 + b)$. The value of $a + b$ is _____.

Lesson 5: Solving Quadratic Equations by Factoring

Assignment

1. Solve.
(a) $(x-8)(x+3)=0$ (b) $(4x+3)(x-6)=0$ (c) $7x(2x-9)=0$ (d) $-3(x+5)(5x-2)=0$

2. Solve.
(a) $x^2+11x=0$ (b) $x^2-144=0$ (c) $49x^2-4=0$

(d) $18x^2=8$ (e) $12x-8x^2=0$ (f) $50-2x^2=0$

3. Solve.
(a) $x^2-9x+14=0$ (b) $x^2+4x-45=0$ (c) $x^2+16x+64=0$

(d) $2x^2+9x-5=0$ (e) $6x^2-11x-10=0$ (f) $24+2x-x^2=0$

4. Write each equation in standard form, then solve.

(a) $3x^2 + 2x = 8$ (b) $10x^2 = 13x + 3$ (c) $x(x - 6) = 27$

(d) $(x + 4)(x - 2) = x + 12$ (e) $(3x + 1)^2 = 16$ (f) $(x - 2)(x + 5) = 4(x - 2)$

(g) $-4x^2 + 20x = 25$ (h) $\frac{1}{3}x^2 + x - 6 = 0$ (i) $8m^2 - 3 = 2m$

5. Solve $5 - 9p - p^2 = p^2$.

6. Consider the function $y = 2x^2 + x - 6$.
(a) Determine the zeros of the function.

(b) State the x-intercepts of its graph and the roots of $2x^2 + x - 6 = 0$.

7. Write a quadratic equation in standard form, with integer coefficients, that has the given roots.
(a) 6 and -2 (b) $-\frac{3}{4}$ and 2 (c) 0 and $\frac{5}{2}$

(d) a double root of $-\frac{2}{3}$ (e) $\frac{7}{3}$ and $-\frac{7}{3}$

8. Write an equation and solve it.
(a) The sum of the squares of two consecutive even integers is 340. Determine the integers.

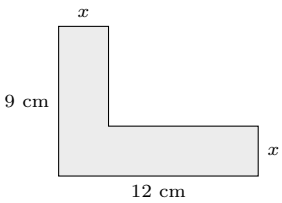
(b) A positive number is multiplied by 3 less than twice the number. The product is 35. Determine the number.

9. The length of a rectangular poster is 4 cm less than three times its width. Its area is 160 cm^2. Determine the dimensions and the perimeter of the poster.

10. The base of a triangle is 6 m shorter than its height. The area of the triangle is 56 m^2. Determine the base and the height.

11. A metal bracket is an L-shape of uniform width x cm. Its outer edges are 12 cm and 9 cm long, as shown.

(a) Show that the area of the bracket is $21x - x^2$ square centimetres.



(b) The area of the bracket is 54 cm^2. Determine x, and explain why one root is rejected.

12. A stone is thrown upward from the top of a 75 m cliff. Its height above the water below is $h = -5t^2 + 10t + 75$ metres after t seconds.

(a) When does the stone hit the water?

(b) At what times is the stone 75 m above the water? Explain what each answer means.

13. How many terms of the arithmetic series 3+7+11+... must be added to give a sum of 300? Use $S_n = \frac{n}{2}[2t_1 + (n-1)d]$ to write a quadratic equation, and solve it.

14. The roots of $3x^2 = 12x$ are

A. 4 only **B.** 0 and 4 **C.** 0 and -4 **D.** 2 and -2

15. A quadratic equation with roots $-\frac{2}{5}$ and 3 is

A. $5x^2 + 13x - 6 = 0$ **B.** $5x^2 - 13x - 6 = 0$ **C.** $5x^2 - 17x + 6 = 0$ **D.** $x^2 - 13x - 6 = 0$

16. The solutions of $(x - 3)(x + 2) = 6$ are

A. 3 and -2 **B.** 9 and 4 **C.** -4 and 3 **D.** 4 and -3

17. Numerical Response The equation $12x^2 + x - 6 = 0$ has roots $x = a$ and $x = -b$, where a and b are positive. The value of $a + b$, to the nearest hundredth, is _____.

18. Numerical Response A polygon with n sides has $\frac{n(n-3)}{2}$ diagonals. A polygon has 54 diagonals. The number of sides is _____.

Lesson 6: Solving Radical Equations and Checking for Extraneous Roots

Assignment

1. State the restriction on the variable.
(a) $\sqrt{x+9}=4$ (b) $\sqrt{5x-3}+1=6$ (c) $\sqrt{12-4x}=2$

(d) $\sqrt{x+5}-\sqrt{2-x}=1$ (e) $\sqrt{3x-1}=\sqrt{10-2x}$ (f) $7-\sqrt{-2x}=3$

2. Solve and verify.
(a) $\sqrt{x-4}=6$ (b) $\sqrt{5x+1}=4$ (c) $\sqrt{2y-7}=1$ (d) $\sqrt{10-3a}=5$

3. Isolate the radical, then solve and verify.
(a) $\sqrt{x+2}-3=4$ (b) $3\sqrt{x-5}=12$ (c) $4\sqrt{2x+1}-5=7$

(d) $10-\sqrt{3x+1}=5$ (e) $\sqrt{\dfrac{x-1}{2}}=3$ (f) $\frac{1}{2}\sqrt{6x}+1=4$

4. Solve and verify. Some equations have no solution: identify any extraneous root.
(a) $\sqrt{x-2}=-5$ (b) $\sqrt{4x+1}+9=6$

(c) $2+3\sqrt{x}=2$ (d) $-\sqrt{x+7}=-2$

5. Solve and verify.
(a) $\sqrt{4x-3} = \sqrt{x+9}$ (b) $\sqrt{7-x} = \sqrt{3x+15}$ (c) $\sqrt{x-8} = \sqrt{3x+2}$

(d) $2\sqrt{x-3} = \sqrt{x+6}$ (e) $\sqrt{5x+4} - 3\sqrt{x} = 0$

6. Each student made an error. Explain the error and give the correct solution.
(a) Ana: $\sqrt{x+6} + 4 = 1 \Rightarrow \sqrt{x+6} = -3 \Rightarrow x + 6 = 9 \Rightarrow x = 3$.

(b) Ben: $2\sqrt{x-1} = \sqrt{x+5} \Rightarrow 2(x-1) = x + 5 \Rightarrow x = 7$.

7. Without solving, explain why each equation has no solution. (a) $\sqrt{2x+5} = -1$ (b) $\sqrt{x-4} + \sqrt{1-x} = 3$

8. Consider $\sqrt{x} = \sqrt{17-3x}$.
(a) State the restriction on x.

(b) Describe how to solve the equation with the intersect feature of a graphing calculator, and state the root.

(c) Verify the root algebraically.

9. The speed of a tsunami wave in the open ocean is approximately $s = \sqrt{9.8d}$, where s is in metres per second and d is the depth of the water in metres. At what depth does a tsunami travel at 140 m/s?

10. Use the pendulum formula $T = 2\pi\sqrt{\dfrac{L}{9.8}}$ (T in seconds, L in metres).
(a) A clock pendulum has a period of 2 s. Determine its length to the nearest centimetre.

(b) Determine the period of a 40 cm pendulum, to the nearest hundredth of a second.

11. The equation $\sqrt{2x+k}=5$ has the root $x=8$. Determine k.

12. The restriction on the variable in $\sqrt{8-2x}$ is

A. $x \geq 4$ **B.** $x \leq 4$ **C.** $x \geq -4$ **D.** $x \leq -4$

13. The solution of $\sqrt{x+5}+6=2$ is

A. $x=11$ **B.** $x=-9$ **C.** $x=-1$ **D.** no solution

14. When solved by squaring, which equation produces an extraneous root?

A. $\sqrt{x-1}=3$ **B.** $\sqrt{2x}=\sqrt{x+4}$ **C.** $\sqrt{x-5}=\sqrt{1-x}$ **D.** $\sqrt{x+1}+2=5$

15. Numerical Response The root of $3\sqrt{2x-5}+1=13$ is $x=$ _____.

16. Numerical Response An object dropped from a height of h metres hits the ground at a speed of $v=\sqrt{19.6h}$ m/s. To hit the ground at 21 m/s, it must be dropped from a height of _____ m.

Lesson 7: Radical Equations That Lead to Quadratics

Assignment

1. Expand and simplify. Assume each radicand is non-negative.

(a) $(x-7)^2$ (b) $(3x-2)^2$ (c) $(5-2x)^2$

(d) $(4+\sqrt{x})^2$ (e) $(2\sqrt{x}-3)^2$ (f) $\left(\sqrt{x-2}+1\right)^2$

2. State the restrictions on x for each equation.

(a) $\sqrt{x+9}=x-3$ (b) $\sqrt{8-x}=x+4$ (c) $\sqrt{2x-1}=3x$ (d) $x+\sqrt{x+5}=1$

3. A student solved $\sqrt{x+4}=x-2$ as shown.

$$x+4=x^2+4 \quad \Rightarrow \quad x^2-x=0 \quad \Rightarrow \quad x(x-1)=0 \quad \Rightarrow \quad x=0 \text{ or } x=1$$

(a) Explain the error.

(b) Solve the equation correctly and verify.

4. Solve and verify. Identify any extraneous roots.

(a) $\sqrt{x+42} = x$

(b) $\sqrt{5x+6} = x$

(c) $x = \sqrt{10-3x}$

(d) $\sqrt{3x+10} = x+2$

5. Isolate the radical, then solve and verify.

(a) $\sqrt{x+7}+5 = x$

(b) $x - \sqrt{2x+6} = 1$

(c) $3\sqrt{x-1} = x+1$

(d) $2x - \sqrt{2x+1} = 1$

6. Solve and verify. (The radicand contains x^2, but the steps are the same.)

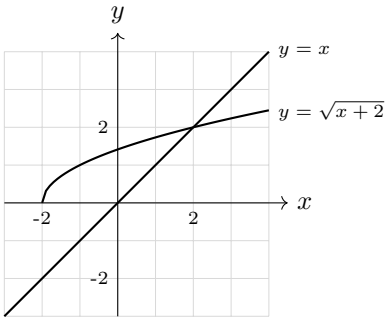
(a) $\sqrt{x^2-5x} = 6$

(b) $\sqrt{2x^2-7} = x+1$

(c) $\sqrt{x^2-8} = \sqrt{2x}$

7. The graphs of $y = \sqrt{x+2}$ and $y = x$ are shown.

(a) Solve $\sqrt{x+2} = x$ algebraically.



(b) Use the graph to explain why one of your answers in (a) is extraneous.

8. Solve each equation. Use either method from Section 4.

(a) $x + \sqrt{x} = 20$

(b) $2x - 7\sqrt{x} + 3 = 0$

9. Solve by squaring twice. Verify each root.

(a) $\sqrt{x+9} - \sqrt{x} = 1$

(b) $\sqrt{x+1} + \sqrt{x-4} = 5$

(c) $\sqrt{2x+3} - \sqrt{x+1} = 1$

(d) $\sqrt{x+2} - \sqrt{2x-3} = 1$

10. When the square root of a number is subtracted from the square root of seven less than twice the number, the result is 1.

(a) Write a radical equation to represent this information.

(b) Solve the equation to determine the number, and state the extraneous root.

11. Twice a number is 3 more than the square root of the number. Write a radical equation and solve it to determine the number.

12. The distance between points (x_1, y_1) and (x_2, y_2) is $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Point A is $(-1, 3)$ and point B is $(k, 2k)$. The distance AB is 5.

(a) Write a radical equation and solve it for k.

(b) Verify both answers and state the possible coordinates of B.

13. When solving $\sqrt{x + 6} = x - 6$, the extraneous root is

A. 3 **B.** 10 **C.** -3 **D.** -10

14. All the restrictions on x in the equation $\sqrt{12 - x} = x$ are

A. $x \leq 12$ **B.** $x \geq 0$ **C.** $0 \leq x \leq 12$ **D.** $x \geq 12$

15. Without solving, which statement about $\sqrt{x - 4} + \sqrt{1 - x} = 2$ is true?

A. It has exactly one root.
B. It has two roots.
C. It has no roots, because no value of x meets both restrictions.
D. It has no roots, because the right side is positive.

16. Numerical Response The root of the equation $\sqrt{6x+1}+1=2x$, to the nearest tenth, is _____.

17. Numerical Response The equation $x-5\sqrt{x}+6=0$ has two roots. Their sum is _____.

Lesson 8: Unit 4 Review

Assignment

1. Factor fully. If it cannot be factored, say so.

(a) $-8x^2 + 20x$

(b) $7y(y + 5) - 2(y + 5)$

(c) $x^3 - 2x^2 + 5x - 10$

(d) $a^2 - 4ab - 21b^2$

(e) $2x^2 + 6x - 56$

(f) $x^2 + 5x + 8$

2. Factor fully.

(a) $12x^2 - 7x - 10$

(b) $-3x^2 + 10x + 8$

(c) $8x^2 + 10xy - 3y^2$

(d) $\frac{1}{2}x^2 + 2x - 6$

(e) $25x^2 - 30x + 9$

(f) $18x^3 - 3x^2 - 6x$

3. Factor fully using substitution.

(a) $(x + 3)^2 - 2(x + 3) - 15$

(b) $2(2x - 1)^2 - 7(2x - 1) + 3$

(c) $x^4 - 10x^2 + 9$

(d) $4x^4 + 11x^2 - 3$

4. Factor fully. If it cannot be factored, say so.

(a) $49x^2 - 64y^2$

(b) $50a^2 - 8b^2$

(c) $x^4 - 16y^4$

(d) $\frac{x^2}{25} - \frac{1}{9}$

(e) $(2x+1)^2 - 36$

(f) $9(x-2)^2 - 4(x+1)^2$

(g) $4x^2 + 25$

5. Solve by factoring.

(a) $x^2 - 11x + 24 = 0$

(b) $10x^2 + 3x = 4$

(c) $3x^2 = 27x$

(d) $(x+2)(x-3) = 4(x-3)$

(e) $9x^2 - 12x + 4 = 0$

6. (a) Write a quadratic equation in standard form whose roots are -3 and 7.

(b) Write a quadratic equation with integer coefficients whose roots are $\frac{3}{4}$ and -2.

(c) Determine the zeros of $y = 2x^2 + 5x - 3$ and the x-intercepts of its graph. Explain how they are related to the roots of $2x^2 + 5x - 3 = 0$.

7. The product of two consecutive odd integers is 143. Write and solve an equation to determine the integers.

8. A ball is thrown upward from the top of a building. Its height, h metres, after t seconds is $h = -5t^2 + 20t + 25$.
(a) How tall is the building?

(b) When does the ball hit the ground?

(c) At what times is the ball 40 m above the ground? Explain why there are two answers.

9. State the restrictions on the variable.
(a) $\sqrt{3x-6} = 4$ (b) $\sqrt{5-x} + \sqrt{x+1} = 3$ (c) $\sqrt{x+6} = x$

10. Solve and verify.
(a) $\sqrt{4x+1} = 7$ (b) $5 - 2\sqrt{x-3} = 1$

(c) $\sqrt{3x-2} = \sqrt{x+8}$ (d) $\sqrt{x+2} + 6 = 3$

11. Solve and verify. Identify any extraneous roots.
(a) $\sqrt{5x+1} = x+1$ (b) $x - \sqrt{x+9} = 3$ (c) $\sqrt{x+6} - \sqrt{x-6} = 2$

12. The time t, in seconds, for an object dropped from a height of h metres to reach the ground is $t = \sqrt{\dfrac{h}{4.9}}$.
(a) How long does an object dropped from 19.6 m take to fall?

(b) A stone dropped from a bridge takes 3 s to reach the water. How high is the bridge?

13. A number decreased by three times its square root is 10. Write a radical equation and determine the number.

14. (a) Ava factored $2x^2 - 8$ as $(2x - 4)(x + 2)$. Her product is correct, but the question was marked wrong. Explain why and factor fully.

(b) Ben solved $x^2 = 5x$ by dividing both sides by x and got $x = 5$. Explain his error and give the full solution.

15. One factor of $9x^2 - 49y^2$ is

A. $9x - 7y$ **B.** $3x - 49y$ **C.** $3x + 7y$ **D.** $9x + 49y$

16. One factor of $6x^2 - x - 15$ is

A. $2x - 3$ **B.** $3x + 5$ **C.** $3x - 5$ **D.** $6x + 5$

17. Numerical Response The expression $12x^2 + 7x - 10$ can be written in the form $(ax - b)(cx + d)$, where a, b, c and d are positive integers. Write the value of a in the first box, b in the second box, c in the third box and d in the fourth box.

18. When factored, $x^2 - 2xy - 24y^2$ and $x^2 + 9xy + 20y^2$ have one binomial factor in common. This factor is

A. $x - 6y$ **B.** $x + 4y$ **C.** $x + 5y$ **D.** $x - 4y$

19. Which of the following is **not** a factor of $x^4 - 17x^2 + 16$?

A. $x + 1$ **B.** $x - 4$ **C.** $x + 2$ **D.** $x^2 - 16$

20. The solution of $16x^2 - 25 = 0$ is

A. $x = \frac{5}{4}$ only **B.** $x = \pm\frac{4}{5}$ **C.** $x = \pm\frac{5}{4}$ **D.** $x = \pm\frac{25}{16}$

21. Numerical Response In order for $25x^2 - kx + 16$ to be a perfect square trinomial, the positive value of k must be _____.

22. The factored form of $4a^2 - (b - 3)^2$ is

A. $(2a - b - 3)(2a + b + 3)$

B. $(2a - b + 3)(2a + b - 3)$

C. $(2a - b + 3)(2a + b + 3)$

D. $(4a - b + 3)(4a + b - 3)$

23. The extraneous root of the equation $x + 2 = \sqrt{x + 14}$ is

A. -5 **B.** -2 **C.** 2 **D.** 5

24. Numerical Response The expression $6(x+1)^2 + 7(x+1) - 3$ can be written in factored form as $(3x + K)(2x + L)$. The value of KL is _____.

25. All the restrictions on x in the equation $\sqrt{x - 3} = \sqrt{7 - x} + 2$ are

A. $x \geq 3$ **B.** $x \leq 7$ **C.** $3 \leq x \leq 7$ **D.** $x \geq 7$

26. The roots of $2 + 5x - 12x^2 = 0$ are

A. $\frac{1}{4}, -\frac{2}{3}$ **B.** $-\frac{1}{4}, \frac{2}{3}$ **C.** $4, -\frac{3}{2}$ **D.** $-4, \frac{3}{2}$

27. Numerical Response The solution of $\sqrt{x + 1} + \sqrt{x - 6} = 7$ is $x =$ _____.
